

Spatial Filtering

INFORMATIKA UNS – PENGOLAHAN CITRA DIGITAL GANJIL 2026/2027

Dasar-dasar Spatial Filtering

Dasar-dasar Spatial Filtering

Spatial Filtering

- The name **filter** is borrowed from frequency domain processing where “**filtering**” refers to **passing**, **modifying**, or **rejecting** specified frequency components of an image.
- **Spatial filtering** modifies an image by **replacing the value of each pixel** by a function of the values of the pixel and its neighbors.
- If the operation performed on the image pixels is linear, then the filter is called a **linear spatial filter**. Otherwise, the filter is a **nonlinear spatial filter**.

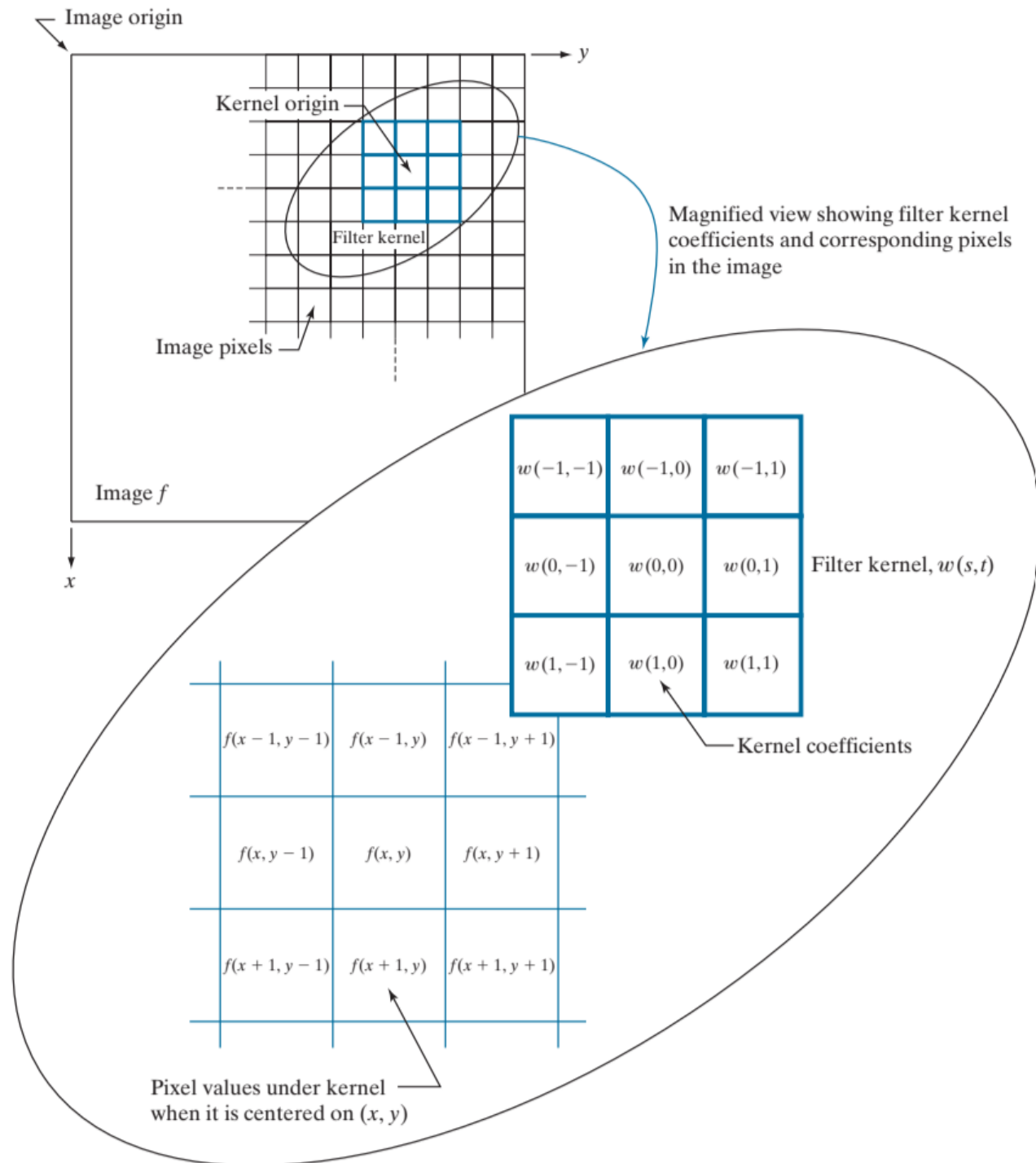
Dasar-dasar Spatial Filtering

Mekanisme Linear Spatial Filtering

- A **linear spatial filter** performs a **sum-of-products** operation between an **image f** and a **filter kernel, w** .

FIGURE 3.28

The mechanics of linear spatial filtering using a 3×3 kernel. The pixels are shown as squares to simplify the graphics. Note that the origin of the image is at the top left, but the origin of the kernel is at its center. Placing the origin at the center of spatially symmetric kernels simplifies writing expressions for linear filtering.



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Mekanisme Linear Spatial Filtering

- Figure 3.28 illustrates the mechanics of linear spatial filtering using a 3×3 kernel. At any point (x, y) in the image, the response, $g(x, y)$, of the filter is the sum of products of the kernel coefficients and the image pixels encompassed by the kernel:

$$g(x, y) = w(-1, -1)f(x - 1, y - 1) + w(-1, 0)f(x - 1, y) + \dots \\ + w(0, 0)f(x, y) + \dots + w(1, 1)f(x + 1, y + 1) \quad (3-30)$$

- As coordinates x and y are varied, the center of the kernel moves from pixel to pixel, generating the filtered image, g , in the process.

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Mekanisme Linear Spatial Filtering

- Observe that the **center** coefficient of the kernel, $w(0, 0)$, aligns with the pixel at location (x, y) .
- For a kernel of size $m \times n$, we assume that $m = 2a + 1$ and $n = 2b + 1$, where a and b are **nonnegative integers**.
- This means that our focus is on kernels of **odd size** in both coordinate directions.
- In general, **linear spatial filtering** of an image of size $M \times N$ with a kernel of size $m \times n$ is given by the expression:

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t) \quad (3-31)$$

- where x and y are varied so that the **center (origin) of the kernel** visits **every pixel** in f once.
- For a fixed value of (x, y) , Eq. (3-31) implements the **sum of products** of the form shown in Eq. (3-30), but for a kernel of arbitrary **odd size**.
- This equation is a central tool in **linear filtering**.

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Spatial Correlation dan Spatial Convolution

- **Spatial correlation** is illustrated graphically in Fig. 3.28. The **correlation** of a **kernel** w of size $m \times n$ with an **image** $f(x, y)$, denoted as $(w \star f)(x, y)$, is given by Eq. (3-31), which we repeat here for convenience:

$$(w \star f)(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t) \quad (3-34)$$

- **Correlation** consists of moving the **center of a kernel** over an image, and computing the **sum** of **products** at each location.

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Spatial Correlation dan Spatial Convolution

- The mechanics of **spatial convolution** are the **same**, except that the correlation kernel is **rotated by 180°**.
- The **convolution** of a kernel w of size $m \times n$ with an image $f(x, y)$, denoted by $(w \star f)(x, y)$, is defined as:

$$(w \star f)(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x - s, y - t) \quad (3-35)$$

- where the **minus signs** align the coordinates of f and w when one of the functions is **rotated by 180°**.

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Spatial Correlation dan Spatial Convolution

- Pada correlation/convolution, terdapat masalah pada koefisien yang tidak dapat diposisikan pada citra. Diantara solusinya adalah menggunakan **padding** (biasanya bernilai 0, atau disebut **zero padding**).
- Contoh padding dan perhitungan correlation dengan kernel 3 x 3:

0	0	0	0	0	0	0
0	60	113	56	139	85	0
0	73	121	54	84	128	0
0	131	99	70	129	127	0
0	80	57	115	69	134	0
0	104	126	123	95	130	0
0	0	0	0	0	0	0

Kernel

0	-1	0
-1	5	-1
0	-1	0

3 x 3

114				

Dari mana dapat **114**?
Slide selanjutnya...

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Spatial Correlation dan Spatial Convolution

- Kernel berukuran $m \times n = 3 \times 3$. Lalu kita mencari nilai a dan b untuk dimasukkan ke Eq. (3-34).
- $m = 2a + 1 \rightarrow 3 = 2a + 1 \rightarrow a = 1$; $n = 2b + 1 \rightarrow 3 = 2b + 1 \rightarrow b = 1$.
- Menghitung correlation memakai Eq. (3-34):

$$(w \star f)(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

$$(w \star f)(x, y) = \sum_{s=-1}^1 \sum_{t=-1}^1 w(s, t) f(x + s, y + t)$$

$$(w \star f)(x, y) = w(-1, -1)f(x - 1, y - 1) + w(-1, 0)f(x - 1, y) + w(-1, 1)f(x - 1, y + 1) + \\ w(0, -1)f(x, y - 1) + w(0, 0)f(x, y) + w(0, 1)f(x, y + 1) + \\ w(1, -1)f(x + 1, y - 1) + w(1, 0)f(x + 1, y) + w(1, 1)f(x + 1, y + 1)$$

$$(w \star f)(x, y) = (0)(0) + (-1)(0) + (0)(0) + \\ (-1)(0) + (5)(60) + (-1)(113) + \\ (0)(0) + (-1)(73) + (0)(121) = 114$$

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Spatial Correlation dan Spatial Convolution

- If the **kernel** and **image** are of sizes $m \times n$ and $M \times N$, respectively, the padding would have to increase to $(m - 1)$ padding elements **above and below** the image, and $(n - 1)$ elements to the **left and right**.
- Under these conditions, the size of the resulting **full correlation** or **full convolution** array will be of size $S_v \times S_h$, where:

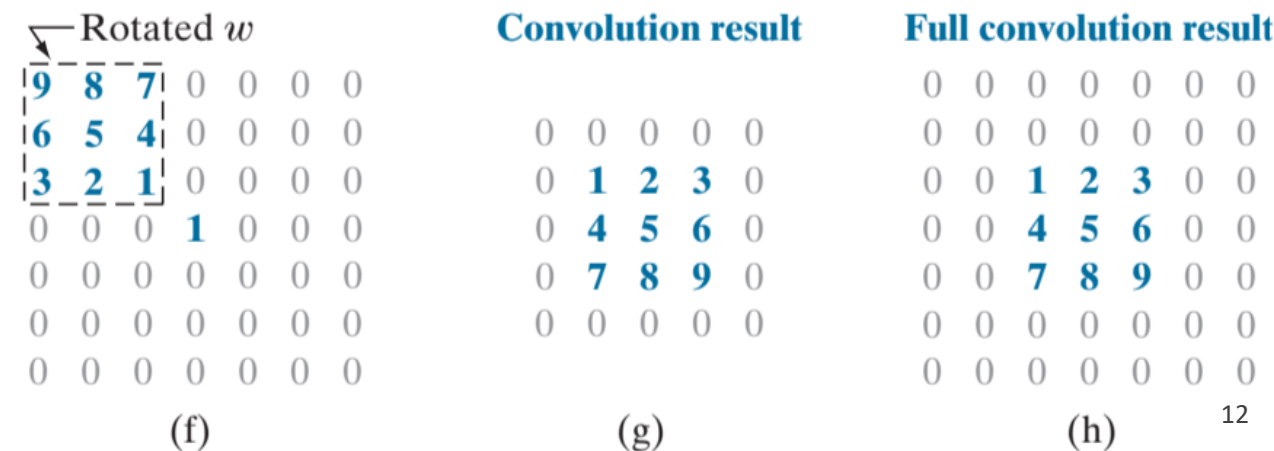
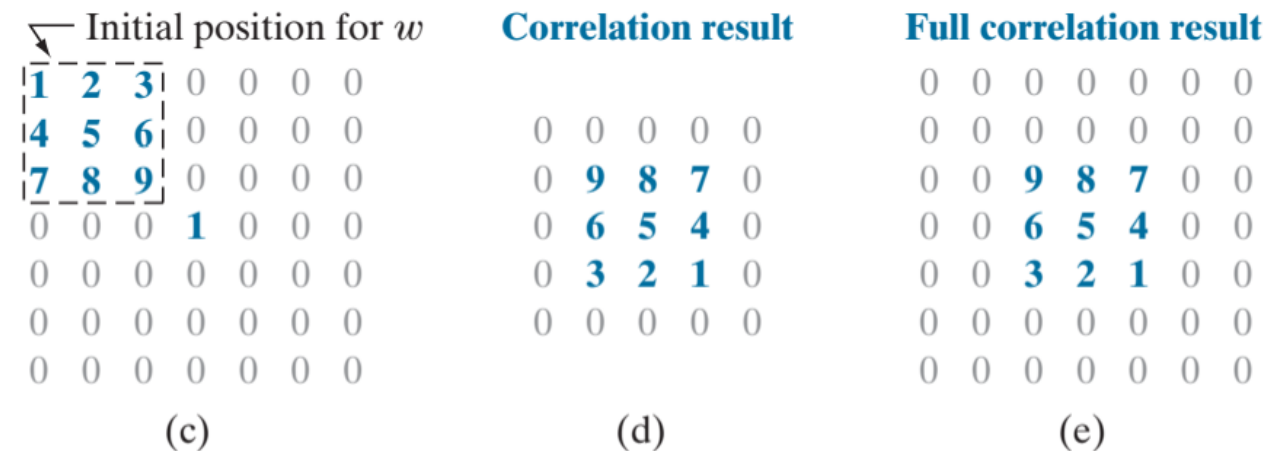
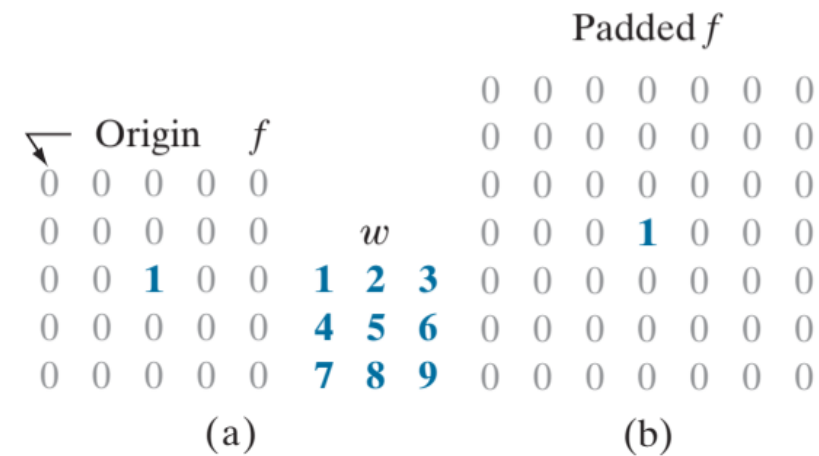
$$S_v = m + M - 1 \quad (3-36)$$

and

$$S_h = n + N - 1 \quad (3-37)$$

Spatial Correlation dan Spatial Convolution

FIGURE 3.30 Correlation (middle row) and convolution (last row) of a 2-D kernel with an image consisting of a discrete unit impulse. The 0's are shown in gray to simplify visual analysis. Note that correlation and convolution are functions of x and y . As these variable change, they *displace* one function with respect to the other. See the discussion of Eqs. (3-36) and (3-37) regarding full correlation and convolution.



Dasar-dasar Spatial Filtering

Spatial Correlation dan Spatial Convolution

- **PERHATIAN!!!**
- **Framework *deep learning*** saat ini menggunakan terminologi “convolution” untuk mendefinisikan operasi “correlation.” Dengan kata lain, **convolution = correlation**.

d2l.ai/chapter_convolutional-neural-networks/conv-layer.html

7. Convolutional Neural Networks > 7.2. Convolutions for Images

7.2.1. The Cross-Correlation Operation

Recall that strictly speaking, convolutional layers are a misnomer, since the operations they express are more accurately described as cross-correlations. Based on our descriptions of convolutional layers in [Section 7.1](#), in such a layer, an input tensor and a kernel tensor are combined to produce an output tensor through a cross-correlation operation.

Referensi: https://d2l.ai/chapter_convolutional-neural-networks/conv-layer.html

Dasar-dasar Spatial Filtering

Separable Filter Kernels

- A 2-D function $G(x, y)$ is said to be **separable** if it can be written as the product of two 1-D functions, $G_1(x)$ and $G_2(y)$; that is $G(x, y) = G_1(x)G_2(y)$.
- A spatial filter kernel is a matrix, and a separable kernel is a matrix that can be expressed as the outer product of two vectors. For example, the 2×3 kernel:

$$w = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

- is **separable** because it can be expressed as the **outer product** of the vectors:

$$\mathbf{c} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{r} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \Rightarrow \quad \mathbf{c} \mathbf{r}^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = w$$

Dasar-dasar Spatial Filtering

Separable Filter Kernels

- A separable kernel of size $m \times n$ can be expressed as the **outer product** of two vectors, $\mathbf{v}$ and $\mathbf{w}$:

$$\mathbf{w} = \mathbf{v}\mathbf{w}^T \quad (3-41)$$

○ where $\mathbf{v}$ and $\mathbf{w}$ are vectors of size $m \times 1$ and $n \times 1$, respectively

- For a **square kernel** of size $m \times m$, we write:

$$\mathbf{w} = \mathbf{v}\mathbf{v}^T \quad (3-42)$$

Dasar-dasar Spatial Filtering

Separable Filter Kernels

- The importance of separable kernels lies in the computational advantages that result from the **associative** property of **convolution**.
- If we have a kernel w that can be decomposed into two simpler kernels, such that $w = w_1 \star w_2$, then it follows from the **commutative** and **associative** properties in Table 3.5 that:

$$w \star f = (w_1 \star w_2) \star f = (w_2 \star w_1) \star f = w_2 \star (w_1 \star f) = (w_1 \star f) \star w_2 \quad (3-43)$$

TABLE 3.5

Some fundamental properties of convolution and correlation. A dash means that the property does not hold.

Property	Convolution	Correlation
Commutative	$f \star g = g \star f$	—
Associative	$f \star (g \star h) = (f \star g) \star h$	—
Distributive	$f \star (g + h) = (f \star g) + (f \star h)$	$f \star (g + h) = (f \star g) + (f \star h)$

Smoothing (Lowpass) Spatial Filters

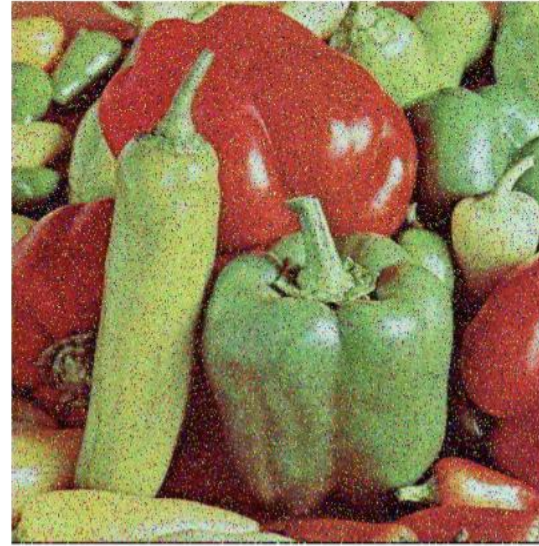
Smoothing (Lowpass) Spatial Filters

Apa itu smoothing?

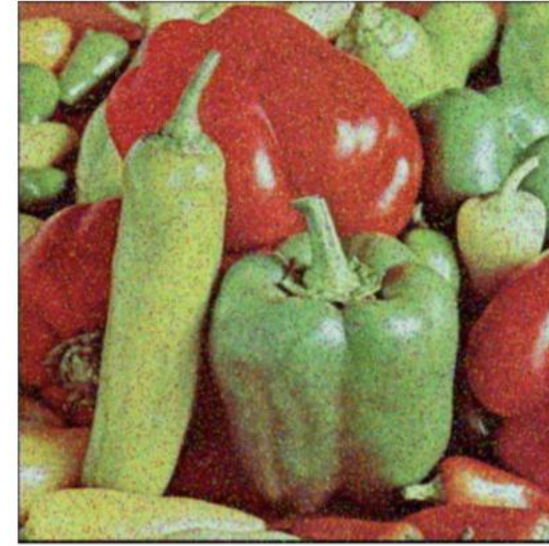
- **Smoothing** (also called **averaging**) spatial filters are used to **reduce sharp transitions** in intensity.
- An application of smoothing is **noise reduction**.
- Smoothing is used to **reduce irrelevant detail** in an image, where “irrelevant” refers to pixel regions that are small with respect to the size of the filter kernel.



Original image



Noisy image



Filtered image

11

- We discuss in this section **lowpass filters** based on **box** and **Gaussian kernels**, both of which are **separable**.

Smoothing (Lowpass) Spatial Filters

Box Filter Kernels

- The simplest, **separable** lowpass filter kernel is the **box kernel**, whose coefficients have the **same value** (typically **1**).
- Box filter menghitung **rata-rata (average)** dari setiap piksel.
- An $m \times n$ box filter is an $m \times n$ array of **1**'s, with a **normalizing constant in front**, whose value is 1 divided by the sum of the values of the coefficients (i.e., $1/(mn)$ when all the coefficients are **1**'s).
- Contoh **3 x 3** box filter:

$$mn = 3 \times 3 = 9$$

$\frac{1}{9} \times$

1	1	1
1	1	1
1	1	1

Smoothing (Lowpass) Spatial Filters

Box Filter Kernels

- [LATIHAN]

- Diberikan citra 8-bit dengan ukuran 7×7 sebagai berikut:

0	0	0	0	9	18	27
0	18	9	180	27	9	108
9	18	27	180	27	9	180
27	252	27	9	180	90	54
0	0	0	0	216	0	81
0	252	108	0	0	0	81
27	180	54	81	108	0	0

- Carilah nilai baru dari **piksel yang ditandai warna merah** jika dilakukan filtering dengan 3×3 box filter!

Smoothing (Lowpass) Spatial Filters

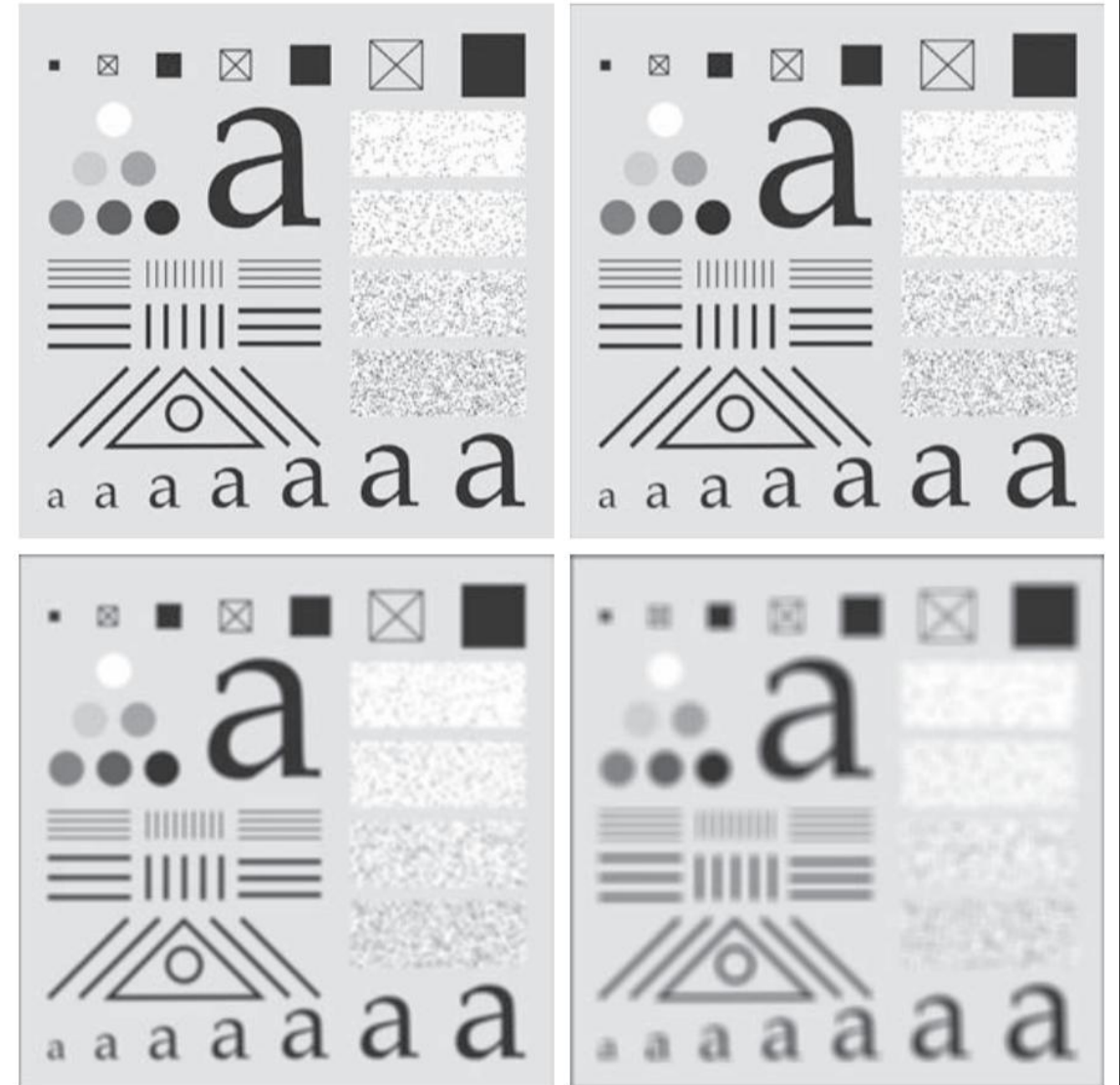
Box Filter Kernels

a b
c d

FIGURE 3.33

(a) Test pattern of size 1024×1024 pixels.

(b)-(d) Results of lowpass filtering with box kernels of sizes 3×3 , 11×11 , and 21×21 , respectively.



Perhatikan **trade-off**: **noise** vs **blur** dan kehilangan detail.

Smoothing (Lowpass) Spatial Filters

Lowpass Gaussian Filter Kernels

- Because of their simplicity, **box filters** are suitable for **quick experimentation** and they often yield smoothing results that are visually acceptable.
- **Box filters** have **limitations**. For example, a defocused lens is often modeled as a lowpass filter, but box filters are poor approximations to the **blurring** characteristics of lenses.
- Another limitation is the fact that **box filters** favor **blurring** along perpendicular directions. In applications involving images with a **high level of detail**, or with **strong geometrical components**, the directionality of box filters often produces **undesirable results**.

Smoothing (Lowpass) Spatial Filters

Lowpass Gaussian Filter Kernels

- Gaussian kernels:

$$w(s, t) = G(s, t) = Ke^{-\frac{s^2 + t^2}{2\sigma^2}} \quad (3-45)$$

- By letting $r = [s^2 + t^2]^{1/2}$ we can write Eq. (3-45) as:

$$G(r) = Ke^{-\frac{r^2}{2\sigma^2}} \quad (3-46)$$

- Variable r is the distance from the center to any point on function G .

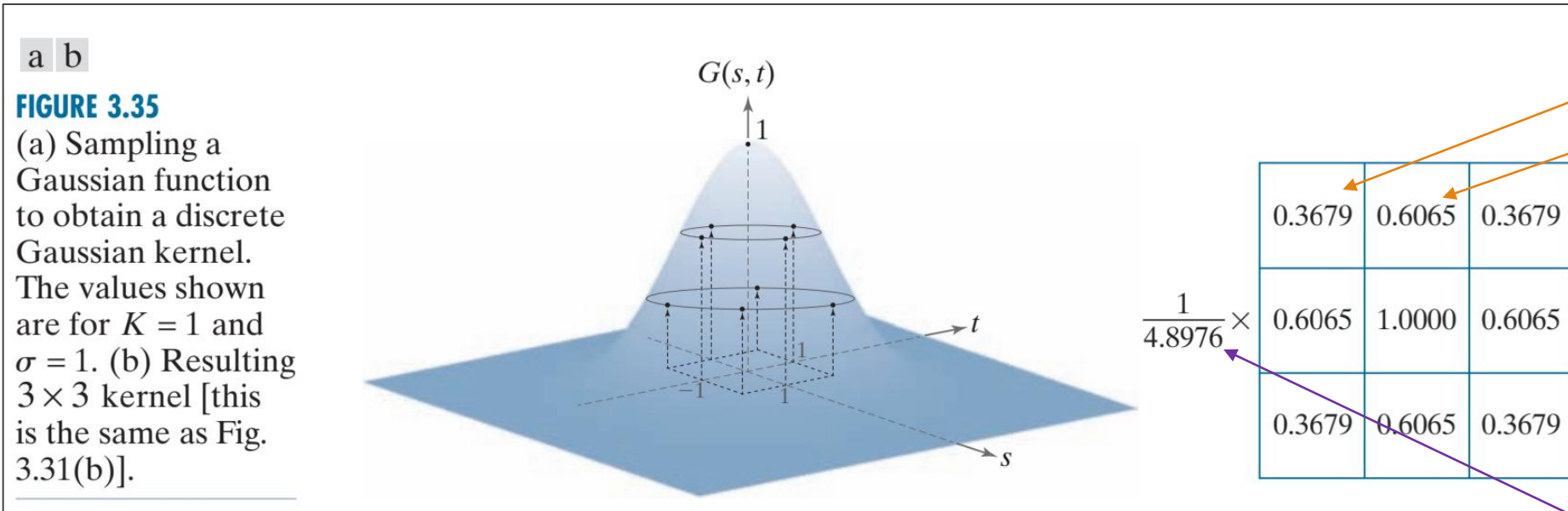
Catatan pada buku: Our interest here is strictly on the bell shape of the Gaussian function; we dispense with the traditional multiplier of the Gaussian PDF (Probability Density Function) and use a general constant, K , instead.

$$\left(\frac{1}{2\pi\sigma^2} \right) e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

Smoothing (Lowpass) Spatial Filters

Lowpass Gaussian Filter Kernels

- Misalkan kita ingin mencari Gaussian filter 3×3 dengan $K = 1$ dan $\sigma = 1$:



(s, t)	r^2	$(1)e^{-\frac{r^2}{2(1)^2}}$
$(-1, -1)$	2	0.3679
$(-1, 0)$	1	0.6065
$(-1, 1)$	2	0.3679
$(0, -1)$	1	0.6065
$(0, 0)$	0	1.0000
$(0, 1)$	1	0.6065
$(1, -1)$	2	0.3679
$(1, 0)$	1	0.6065
$(1, 1)$	2	0.3679
Total:		4.8976

Smoothing (Lowpass) Spatial Filters

Lowpass Gaussian Filter Kernels

- Figure 3.36(b) shows the result of lowpass filtering the test pattern with this kernel. Comparing this result with Fig. 3.33(d), we see that the Gaussian kernel resulted in significantly **less blurring**.
- Fig. 3.36(b) dan Fig. 3.33(d) sama-sama menggunakan kernel berukuran 21×21 .



a b c

FIGURE 3.36 (a) A test pattern of size 1024×1024 . (b) Result of lowpass filtering the pattern with a Gaussian kernel of size 21×21 , with standard deviations $\sigma = 3.5$. (c) Result of using a kernel of size 43×43 , with $\sigma = 7$. This result is comparable to Fig. 3.33(d). We used $K = 1$ in all cases.

Smoothing (Lowpass) Spatial Filters

Order-Statistic (Nonlinear) Filters

- Sebelumnya kita sudah membahas **linear spatial filter**.
- Order-statistic filters are **nonlinear** spatial filters whose response is based on ordering (ranking) the pixels contained in the region encompassed by the filter.
- Smoothing is achieved by replacing the value of the center pixel with the value determined by the ranking result.
- The best-known filter in this category is the **median filter**.

Smoothing (Lowpass) Spatial Filters

Order-Statistic (Nonlinear) Filters

- **Median filters** provide excellent **noise reduction** capabilities for certain types of random noise, with considerably **less blurring** than **linear** smoothing filters of **similar size**.
- **Median filters** are particularly effective in the presence of **impulse noise** (sometimes called **salt-and-pepper noise**, when it manifests itself as **white** and **black dots** superimposed on an image).
- Selain **median filter**, contoh lain dari order-statistic (nonlinear) filter adalah **min filter** dan **max filter**.

Smoothing (Lowpass) Spatial Filters

Order-Statistic (Nonlinear) Filters

- Contoh penerapan median filter 3×3 ; misal piksel yang di tengah (35) terkena *noise* (derau) dan ingin diproses menggunakan median filter 3×3 :

13	10	15	14	18
12	10	10	10	15
11	11	35	10	10
13	9	12	10	12
13	12	9	8	10

(a) *Pixel* bernilai 35 terkena derau

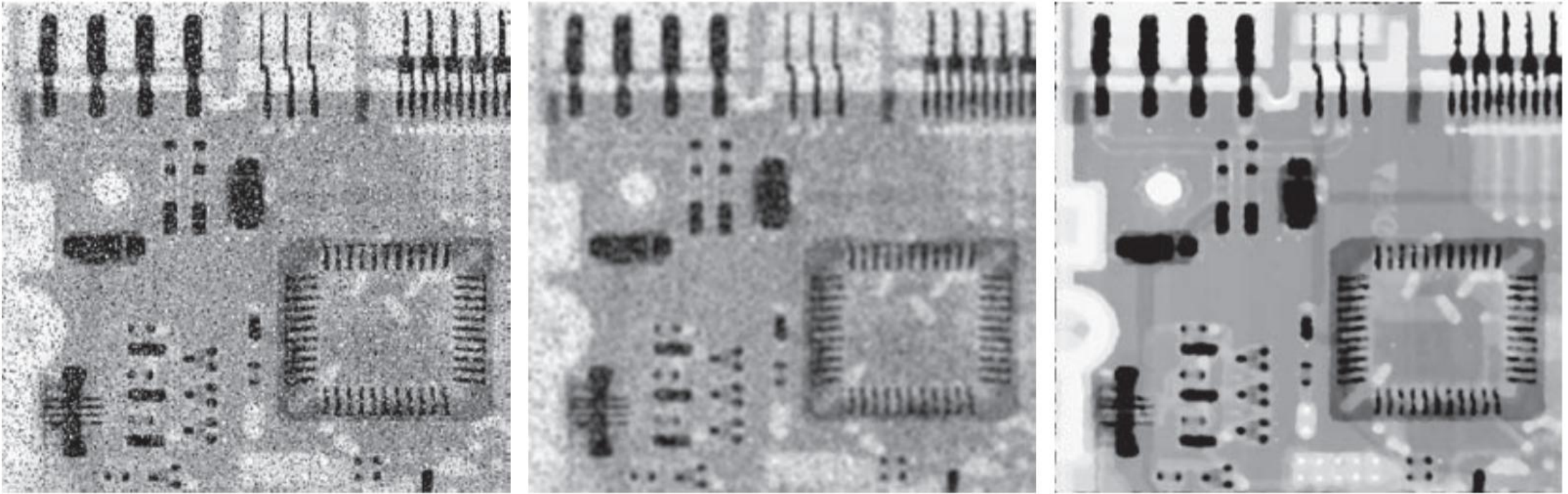
13	10	15	14	18
12	10	10	10	15
11	11	10	10	10
13	9	12	10	12
13	12	9	8	10

(b) 35 diganti dengan median dari kelompok 3×3 *pixel*

- Diurutkan piksel nya dari yang terkecil: 9, 10, 10, 10, 10, 10, 11, 12, 35.
- Didapatkan median = 10 (warna merah).

Smoothing (Lowpass) Spatial Filters

Order-Statistic (Nonlinear) Filters



a b c

FIGURE 3.43 (a) X-ray image of a circuit board, corrupted by salt-and-pepper noise. (b) Noise reduction using a 19×19 Gaussian lowpass filter kernel with $\sigma = 3$. (c) Noise reduction using a 7×7 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Smoothing (Lowpass) Spatial Filters

Order-Statistic (Nonlinear) Filters

- [LATIHAN]

- Diberikan citra 3-bit dengan ukuran 7×7 sebagai berikut:

4	1	3	4	6	2	2
0	1	2	7	1	4	4
5	5	6	6	1	5	5
2	1	5	1	5	0	1
3	2	1	2	4	0	3
0	2	3	7	4	2	2
3	2	5	6	2	3	1

- Carilah nilai baru dari **piksel yang ditandai warna merah** jika dilakukan filtering dengan:
 - a. 3×3 median filter
 - b. 5×5 min filter
 - c. 3×3 max filter

Smoothing (Lowpass) Spatial Filters

Order-Statistic (Nonlinear) Filters

- [LATIHAN]

- Diberikan citra 3-bit dengan ukuran 7×7 sebagai berikut:

4	1	3	4	6	2	2
0	1	2	7	1	4	4
5	5	6	6	1	5	5
2	1	5	1	5	0	1
3	2	1	2	4	0	3
0	2	3	7	4	2	2
3	2	5	6	2	3	1

- Carilah nilai baru dari **piksel yang ditandai warna merah** jika dilakukan filtering dengan 3×3 median filter! Gunakan *zero padding* untuk setiap tepi citra!

Terima Kasih
